

BIO DATA



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1. Educational Qualification

Degree	Branch / Specialization	Institute/University	Year of Passing
B.E.	Computer Science and Engineering	Madurai Kamaraj University, Madurai	1999
M.E	Computer Science and Engineering	Manonmaniam Sundaranar University, Tirunelveli	2011
Ph.D	Information and Communication Engineering	Anna University Chennai	2015

2. Professional/ Industry Experience : (in chronological order)

S.No	Designation/ Department	Institution/ Organization	Period	
			From	To
1	Professor and Head/AI&DS	Ramco Institute of Technology Rajapalayam, Tamilnadu, India	20.10.2021	Till date
2	Professor/CSE	Ramco Institute of Technology Rajapalayam, Tamilnadu, India	02.11.2020	19.10.2021
3	Associate Professor//CSE	Ramco Institute of Technology Rajapalayam, Tamilnadu, India	01.11.2018	01.11.2020
4	Associate Professor/IT	National Engineering College, Kovilpatti	01.08.2016	26.10.2018
5	Assistant Professor(SG)/IT	National Engineering College, Kovilpatti, Tamilnadu, India	06.08.2012	30.07.2016
6	Assistant Professor /IT	National Engineering College, Kovilpatti, Tamilnadu, India	1.11.2011	05.08.2012
7	Lecturer/IT	National Engineering College, Kovilpatti, Tamilnadu, India	01-12-2006	31.10.2011
8	Lecturer/CSE	SACS MAVMM Engg College, Madurai	14.08.2006	30.11.2006
9	Lecturer/CSE	Odaiyappa college of Engg & Tech Theni, Tamilnadu, India	19.06.2006	07.08.2006
9	Lecturer/CSE	Arulmigu Kalasaligam Polytechnic Srivilliputtur	13.08 .2001	12.06.2006

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3. Research Area: Artificial Intelligence, Bigdata Analytics, Ad hoc Networks

4 Publications

4.1 JOURNAL PUBLICATIONS :

1. Pradeepa S, Niveda Gaspar, Vimal.S Subbulakshmi.P,Ahmed Alkhayyat, **Kaliappan.M**, “Classifying Promoters by Interpreting the Hidden Information of DNA Sequences for disease prediction in clinical laboratories using Gaussian Decision Boundary Estimation” Intelligent Decision Technologies, IOS Press, Oct 2023,IF 1.0 , Accepted, SCI
2. Azhagu Jaisudhan Pazhani.A, P. Gunasekaran, Vimal Shanmuganathan, Sangsoon Lim, **Kaliappan Madasamy**, Rajesh Manoharan and Amit Verma, Peer–Peer Communication Using Novel Slice Handover Algorithm for 5GWireless Networks, Journal of Sensor and Actuator Networks, Vol.11, No.82, November,2022. SCI, <https://doi.org/10.3390/jsan11040082>
3. L.Ganesan, C.Umarani, **M.Kaliappan**, VIMAL SHANMUGANATHAN, Seifedine Kadry, Yunyoung Nam, Texture Image Analysis for Larger Lattice Structure using Orthogonal Polynomial framework, Journal of Information Technology and Control, SCI, Vol. 51 No. 3, September 2022, SCI, IF:1.22
4. Karpagarajesh, G., Santhana Krishnan, R., Harold Robinson, S. Vimal, S. Thamizharasan, P. Subbulakshmi, **M. Kaliappan**, Comparative analysis of FSO, OFC and diffused channel links in photonics using artificial intelligence based S-band, C-band and L-band of the attenuation metrics, Optical and Quantum Electronics, Vol.54, No.420, PP.1-20, June 2022. <https://doi.org/10.1007/s11082-022-03762-0>
5. K. Lakshmi Narayanan, R. Santhana Krishnan, Y. Harold Robinson, E. Golden Julie, S. Vimal, V. Saravanan, **M. Kaliappan**, “Banana Plant Disease Classification using Hybrid Convolutional Neural Network”, Computational Intelligence and Neuroscience, Hindawi, Vol.2022, Article ID 9153699, Feb.2022, <https://doi.org/10.1155/2022/9153699>, SCI, IF 3.633.
6. S. Gopikumar, J. Rajesh Banu, Y. Harold Robinson, S. Raja, S. Vimal, Danilo Pelusi & **M Kaliappan**, Geo Spatial Based Real Time Monitoring on Eutrophic Evaluation of Porunai River Basin for Pollution Risk Assessment, European Journal of Remote Sensing, pp:1-14, <https://doi.org/10.1080/22797254.2021.2025152>, Feb 2022,SCIE, IF 3.71.
7. S Vimal, Y Harold Robinson, **M Kaliappan**, Abdullellah A Alaboudi, Ashish Kr Luhach and Noor Zaman, AI based Forecasting of Influenza Patterns from Twitter Information using Random Forest algorithm, Human-centric Computing and Information Sciences , Vol.11, No.33 , pp.1-14,2021, <https://doi.org/10.22967/HGIS.2021.11.033> , SCI, IF:3.7
8. **M. Kaliappan**, S. Vimal, K. Vijayalakshmi; Mi Young Lee, S. Manikandan, OSDY:Embedded system based Object Surveillance Detection System with small drone using Deep YOLO, EURASIP Journal on Image and Video Processing, Article no: 19, pp.2-14, 2021, DOI:<https://doi.org/10.1186/s13640-021-00559-1>
9. **Kaliappan Madasamy**, Vimal Shanmuganathan, Gaurav Dhiman, K.Vijayalakshmi, P.SureshKumar, “Enhancing the QOS of far field networking and communication using the optical properties of graphene” in Materials Today: Proceedings. Online March 2021. (Scopus Indexed), <https://doi.org/10.1016/j.matpr.2021.03.678>
10. S. Vimal,Y. Harold Robinson,M. Kaliappan ,K. Vijayalakshmi,Sanghyun Seo, A method of progression detection for glaucoma using K-means and the GLCM algorithm toward smart medical prediction, Journal of supercomputing, Published online: March 2021, <https://doi.org/10.1007/s11227-021-03757-w>. SCIE, IF 2.6

11. S.Vimal, Y. Harold Robinson, **M.Kaliappan**, Subbulakshmi Pasupathi and A.Suresh, “*Q Learning MDP Approach to Mitigate Jamming Attack Using Stochastic Game Theory Modelling With WQLA in Cognitive Radio Networks*”, Journal of Platform Technology, Vol. 9, No. 1, March 2021.
12. R Venkatesh, C Bala Subramanian, **M Kaliappan**, Rainfall Prediction using Generative Adversarial Networks with Convolution Neural Network, Soft Computing(Springer), Published online January 2021, DOI: 10.1007/s00500-020-05480-9, SCI, IF:3.05
13. Subbulakshmi Pasupathi, Vimal Shanmuganathan, **Kaliappan Madasamy**, Harold Robinson Yesudhas & Mucheol Kim, Trend analysis using agglomerative hierarchical clustering approach for time series big data, The Journal of Supercomputing . Published online January 2021. DOI: <https://doi.org/10.1007/s11227-020-03580-9>, SCI, Impact Factor: 2.469
14. Jackins, V., Vimal, S., **Kaliappan, M.**, Mi Young Lee, AI-based smart prediction of clinical disease using random forest classifier and Naive Bayes. Journal of Supercomputing, Published online Nov 2020. DOI: <https://doi.org/10.1007/s11227-020-03481-x>, SCI, IF.2.4
15. **M. Kaliappan**, A.Niranjana Devi, R.Sandhiya, D.Lakshmanan, S.Selvakumari, Water Tank Monitoring and Flame Detection Application Using LORA, Journal of Green Engineering, Vol.10,No.9, pp. 5748–5762, October 2020. **Scopus Indexed**
16. B. Vijayalakshmi, K. Ramar, **M. Kaliappan** , NZ Jhanjhi, Sahil Verma, K. Vijayalakshmi, S. Vimal, Kavita, Uttam Ghosh, An Attention Based Deep Learning Model For Traffic Flow Prediction Using Spatio Temporal Features Towards Sustainable Smart City, International Journal of Communication Systems, Published online November 2020, DOI:<http://dx.doi.org/10.1002/dac.4609>
17. M. Sivaram, **M.Kaliappan**, S. Jeya Shobana, M. Viju Prakash,V. Porkodi, K.Vijayalakshmi, S.Vimal, A.Suresh, Secure Storage Allocation Scheme using Fuzzy Based Heuristic Algorithm for Cloud, Journal of Ambient Intelligence and Humanized Computing, **IF:1.91**, Published Online 14 May 2020, DOI: 10.1007/s12652-020-02082-z.
18. Ruben Gonzalez Crespo, Vimal S, Manju Khari, L. Kalaivani, Nilanjan Dey, **M Kaliappan** ,Energy enhancement using multi-objective ant colony optimization with double Q learning algorithm for IOT based cognitive radio networks, Computer Communications(Elsevier), **IF:2.76**, Vol. 154, 15, 2020, pp. 481-490., <https://doi.org/10.1016/j.comcom.2020.03.004>
19. R.Venkatesh, C.Bala subramanian **M.Kaliappan** “Development of Big Data Predictive Analytics Model for Disease Prediction using Machine learning Technique”, Journal of Medical Systems (Springer), Vol.43, Article number: 272, 2019, **IF:2.09**, DOI : 10.1007/s10916-019-1398-y.
20. R.Geetha, S.Siva subramanian, **M.Kaliappan**, S.Vimal, Dr.Suresh Annamalai, “Cervical Cancer Identification With Synthetic Minority Oversampling Technique And PCA Analysis Using Random Forest Classifier”, Journal of Medical Systems (Springer), **IF:2.09**, Vol.43, Article number: 286, 2019, DOI: 10.1007/s10916-019-1402-6.
21. Nattar kannan, S. Siva Subramanian, **M. Kaliappan**, S.Vimal, A. Suresh and P. Subbu lakshmi, Artificial Intelligence based attack mitigation Using Stochastic Game Theory Modelling with WQLA in Cognitive Radio Networks”, Journal of Web Engineering, **IF: 0.4**, Vol. 17, No.6, pp.3599 – 3618, 2018

22. S. Vimal, **M. Kaliappan**, A. Suresh, P. Subbu lakshmi, Sanjeev Kumar, and Dinesh Kumar, “Development of Cloud Integrated Internet of Things Based Intruder Detection System”, Journal of Computational Theoretical. Nano science, Vol.15, pp.3570–3575, 2018.
23. S. Vimal, L. Kalaivani, **M. Kaliappan**, A. Suresh, Xiao-Zhi Gao, R. Varatharajan, “Development of secured data transmission using machine learning-based discrete-time partially observed Markov model and energy optimization in cognitive radio networks” Journal of Neural Computing and Applications, **IF:4.66**, Vol.32, pp.151-161,2020, DOI 10.1007/s00521-018-3788-3
24. K. Nattar kannan, S.Sivasubramanian, **M.Kaliappan**, S.Vimal, “Predictive big data analytic on demonetization data using support vector machine”, Cluster Computing (Springer), **IF:1.8**, Vol.22, pp.14709-14720, 2019, DOI: <https://doi.org/10.1007/s10586-018-2384-8>.
25. Suthagar Ilango S.Vimal, **M.Kaliappan**, P. Subbulakshmi “Optimization using artificial bee colony based clustering approach for big data”, Cluster Computing (Springer), **IF: 1.8**, Vol.22, pp. 12169–12177, September 2019, DOI: 10.1007/s10586-017-1571-3.
26. S.Vimal, L. Kalaivani, **M.Kaliappan** “Collaborative approach on mitigating spectrum sensing data hijack attack and dynamic spectrum allocation based on CASG modeling in wireless cognitive radio networks”, Cluster Computing (Springer), **IF:1.8**, Vol.22, pp.10491-10501, 2019 , DOI:10.1007/s10586-017-1092-0, 2019
27. **M.Kaliappan** Dr.B.Paramasivan “Susan Agustine, Enhancing Energy Efficiency and Load Balanced Clustering Technique in MANET using Genetic Algorithms Network and Computer Application” (**Elsevier**) **IF: 6.28**, Vol.73, pp.35-43,2016 DOI:<http://dx.doi.org/10.1016/j.jnca.2016.07.003>.
28. **M. Kaliappan**, E. Mariappan, M. Viju Prakash, and B. Paramasivan “Load Balanced Clustering Technique in MANET using Genetic Algorithms Defence Science Journal **IF:0.58**, Vol. 66, No. 3, 2016, pp.251-258, DOI :10.14429/dsj.66.9205.
29. E. Mariappan , **M. Kaliappan** S. Vimal, “Energy Efficient Routing Protocol Using Grover’s Searching Algorithm for MANET”, Asian Journal of Information Technology Vol.14,no.24, pp.4986-4994,2016, DOI:10.3923/ajit.2016.4986.4994
30. **M.Kaliappan**, Dr.B.Paramasivan, Enhancing secure routing in Mobile Ad Hoc Networks using a Dynamic Bayesian Signalling Game model, Computers & Electrical Engineering (Elsevier) **IF:0.816**, Vol.4, pp.301-313(2015)
31. M.Viju Prakash, **M. Kaliappan**, B. Paramasivan, Energy Efficient Dynamic Load Balanced Clustering Protocol using Memory Enhanced Genetic Scheme and Elitism based Immigrant Genetic Scheme for MANET Journal of Pure and Applied Microbiology **IF:0.07** Vol. 9(Spl. Edn.2), pp. 655-665, Nov.2015
32. Balasubramanian Paramasivan, Maria Johan Viju Prakash, and **Madasamy Kaliappan**, Development of a Secure Routing Protocol using Game Theory Model in Mobile Ad Hoc

Networks, Journal of Communications and Networks, IF:1.0, Vol. 17, No. 1, pp.75-83, February 2015.

33. **M.Kaliappan**, Paramasivan, B, Development of secure cluster based multipath routing scheme for MANET, International Journal of Applied Engineering Research, vol.10, no.12, pp.31565-31583, ISSN: 0973-4562, 2015.
34. **M.Kaliappan** , Dr.B.Paramasivan Secure and Fair Cluster Head Selection Protocol for Enhancing Security in Mobile Ad Hoc Networks, Scientific World Journal, IF:1.2, Volume 2014, Article ID 608984, 6 pages,2014.
35. **M.Kaliappan** Dr.B.Paramasivan K.Mohaideen pitchai K.Bhavaneswari, Enhancement of Security and Congestion Adaptiveness in Dynamic Source Routing Protocol, International Journal of Networking and Communication Engineering, Vol.4,No.7, Pp 266-370,June 2012

4.2 CONFERENCE PUBLICATIONS

1. **Kaliappan Madasamy**, Vimal Shanmuganathan, Nithish, Vishakan, Vijayabhaskar, Muthukumar, Balamurali Ramakrishnan, Ramnath.M, Benign and Malignant Cancer Prediction using Deep Learning and Generating Pathologist Diagnostic Report, Second International Conference on Internet of Things and Health (IoTHIC-2023), October 20-21 2023, Haliç University, Istanbul, Turkey. (Scopus)
2. Karpagavalli. C; **Kaliappan. M**; Amuthachenthiru. K, Radial Basis Function Neural Network for Intrusion Detection and Feed Forward Artificial Neural Network for Attack Mitigation in IoT, 2023 Second International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), 23-25 August 2023, Trichy, India, Publisher: IEEE, DOI: 10.1109/ICAISS58487.2023.10250458.Scopus Indexed
3. K.Vijayalakshmi, **M.Kaliappan**, R.Siva Ranjini, M.Sasi Rekha, and R.Selva Ishwarya, “Automatic Book Reading System for Visually Impaired Person using Deep Learning”, National Web Conference on Challenges and Innovation in Engineering And Technology (NWCCIET-2021), 19-20 March, 2021.
4. **M.Kaliappan**, K.Vijayalakshmi, and P.Rajarajeswari, “Automatic Questionnaire Generator using Deep Learning”, National Web Conference on Challenges and Innovation in Engineering And Technology (NWCCIET-2021), 19-20 March, 2021
5. S. Vimal, Y. Harold Robinson, **M. Kaliappan**, K. Vijayalakshmi, Sanghyun Seo, Progression Detection of Glaucoma using K-Means and GLCM Algorithm, The 22nd International Conference on Artificial Intelligence-(ICAI'20), July 27-30, 2020, Las Vegas, USA.
6. P. Subbulakshmi, S. Vimal, **M. Kaliappan**, Y. Harold Robinson, Mucheol Kim, Trend Analysis using Agglomerative Hierarchical Clustering Approach for Time Series Big Data, The 22nd International Conference on Artificial Intelligence-(ICAI'20), July 27-30, 2020, Las Vegas, USA.
7. V. Jackins, S. Vimal, **M. Kaliappan**, Mi Young Lee, Prediction of Clinical Disease with AI based Multiclass Classification using Naive-Bayes and Random Forest Classifier, The

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22nd International Conference on Artificial Intelligence-(ICAI'20), July 27-30, 2020, Las Vegas, USA.

8. **M.Kaliappan**, C.A Yogaraja , A.Niranjana Devi, R.Sandhiya , and S.Selvakumari, Water Tank Monitoring and Flame Detection using LORA, International Web Conference on Smart Engineering Technologies-2020, 26.06.2020 & 27.06.2020, Location: RAMCO INSTITUTE OF TECHNOLOGY, Rajapalayam
9. R.Venkatesh, **M.Kaliappan**, C. Balasubramanian K.R.Geerthanapriya, Rainfall Prediction System Using Generative Adversarial Networks, International Web Conference on Smart Engineering Technologies-2020, 26.06.2020 & 27.06.2020, Location: RAMCO INSTITUTE OF TECHNOLOGY, Rajapalayam
10. **M. Kaliappan**, B. Vijayalakshmi, Manikandan, M.Shrimalavika, G.Shunmugavalli and A. VijayaMalini, SEO Analysis for B2B websites using Google Webmaster tools, International Web Conference on Smart Engineering Technologies-2020, 26.06.2020 & 27.06.2020, Location: RAMCO INSTITUTE OF TECHNOLOGY, Rajapalayam
11. **M Kaliappan** B Paramasivan T. Dhivya Development of Energy Efficient Dynamic Load Balanced Clustering Protocol Using Dynamic Genetic Algorithm in MANET , National Conf. on Frontiers in Applied Sciences And Computer Technology (FACT'15), Vol. 03, March 06 & 07, 2015.
12. **M Kaliappan** G.Murugeswari, Secure Congestion Adaptive Routing protocol for Mobile Adhoc Network, Intelligent Computing in Communication & Automation (NCICCA-2011), 08.04.2011.
13. B.Paramasivan K. Mohaideen Pitchai, **M.Kaliappan**, A Novel routing protocol for improving energy efficiency using fuzzy logic in wireless sensor networks, IEEE Sensor 2012 ISBN:978-1-4577-1765-9, pp.1388-139,128.10.2012 to 31.10.2012.
14. **M Kaliappan**, B Paramasivan Enhancing Energy Efficiency in Mobile Adhoc Network Using Agglomerative Hierarchical Clustering Technique, ASCENT 2015-Proceedings of Information Systems & Engineering, pp.1-10,12.03.2015-13.03.2015
15. **M.Kaliappan**, Secure Congestion Adaptive Routing protocol for Mobile Adhoc Networks, National Conference on Emerging Trends in Computing Technologies, pp.1-6 23.03.2011
16. S. Rimlon Shibi, **M. Kaliappan**, L. Jerart Julius IPSec Provisioning in WiMAX Networks to Enhance the Security International Conference on Recent Trends in Computational Methods, Communication and Controls, pp.6-12, 04.04.2012.
17. G. Sivakumar, **M.Kaliappan**, L. Jerart Julius, Enhancing the Performance of MANET using EESCP, International Conference on Pattern Recognition, Informatics and Medical Engineering PRIME2012, pp.221-230, 21.03.2012 to 23.03.2012.

18. S. Rimlon Shibi, **M. Kaliappan**, L. Jerart Julius, Analysis and Security Enhancement in Wimax Networks, Emerging Technology Trends in Advanced Engineering Research- ICEN 201, 20.02.2012 to 21.02.2012.
19. S. Rimlon Shibi, **M. Kaliappan**, L. Jerart Julius, Development of Energy Efficiency and Secure Communication Protocol in MANET, Emerging Trends in Computer & Communication Technologies – EMICS'12, 16.02.2012 to 17.02.2012.
20. S.M.R.Sreedhar Surya Prakash, S.Vinoth, **M.Kaliappan**, Energy Efficient Routing Protocol Based On Grover's Searching Algorithm For Mobile Adhoc Networks, National Conference on Recent Trends In Information Technology, 11.04. 2014.
21. J Vivek narayanan, **M Kaliappan**, Automatic Polling Service in Electronic Voting Machine, National Conf. on Frontiers in Applied Sciences And Computer Technology (FACT'15), Vol. 03, March 06 & 07, 2015.
22. P.Bagavath J.Subash H.Venkates **M. Kaliappan**, Optimal Double Route Search for Enhancing Energy Efficiency in Mobile Adhoc Network, National Conference on Recent Issues in Network Galaxy 2014-RING'14, pp.7-13 24.03. 2014.
23. T.Dhivya, **M Kaliappan**, Enhancing energy efficient dynamic load balanced clustering protocol using Dynamic Genetic Algorithm in MANET, International Conference on Innovations in Communications and Computer Engineering- ICICCE'15, pp.48-49, 15.02.2015.
24. **M.Kaliappan** R.Sujitha P.Subbulakshmi A.H.Ragamathu nisa begum, Security Enhancement In MANET Using Game Theory Approach, International Conference On "Advanced Computing, Machines And Embedded Technology", pp.683-690 8.3.2013 & 9.3.2013
25. K. Amsaveni, **M. Kaliappan** P.Subbulakshmi Dr.D.Manimega lai, An Energy Efficient Leader Election Model In MANET Using Mechanism Design Theory, International Conference On "Advanced Computing, Machines And Embedded Technology, pp.691-698 8.3.2013 & 9.3.2013.
26. R.Sujitha, **M.Kaliappan**, P.Subbulakshmi, A Comprehensive Survey on Security Enhancement in Mobile Adhoc Networks using Game theory Approaches. International Conference on Egovernance And Cloud Computing Services-Egov'12. Vol.1, pp.602-606, 17.10.2012 to 18.10.2012.
27. K. Amsaveni, **M.Kaliappan**, P.Subbulakshm, M B.Paramasivan, A Survey On Leader Election Models In Mobile Ad Hoc Networks, International Conference on Egovernance And Cloud Computing Services-Egov'12, Vol.1, pp.607-613 17.10.2012 .

4.3 Book/Book chapter publications

1. Pradeepa Sampath, Vimal Shanmuganathan, JanmenjoyNayak, Prasun Chakrabarti, **Kaliappan Madasamy**, A Knowledge Discovery framework for Coronavirus Disease 2019 (COVID -19) disease from PubMed Abstract using Association Rule Hypergraph (AR-Hypergraph)- Elsevier book chapter Data Science in medical field, Accepted – Scopus- Jan 2024
2. Vimal S., Suresh A., Subbulakshmi P., Pradeepa S., **Kaliappan M.** (2020) Edge Computing-Based Intrusion Detection System for Smart Cities Development Using IoT in Urban Areas. In: Kanagachidambaresan G., Maheswar R., Manikandan V., Ramakrishnan K. (eds) Internet of Things in Smart Technologies for Sustainable Urban Development. EAI/Springer Innovations in Communication and Computing. Springer, Cham , DOI:https://doi.org/10.1007/978-3-030-34328-6_14).
3. S. Vimal, V. Jeyabalaraja, P. Subbulakshmi, A. Suresh, **M. Kaliappan**, S. Koteeswaran, Deep Learning-Based Decision-Making with WoT for Smart City Development, In Aarti Jain, Rubén González Crespo, Manju Khari(eds), Smart Innovation of Web of Things, 1st Edition, Published 2020, CRC Press, Pages12, ISBN:9780429298462, <https://doi.org/10.1201/9780429298462>
4. Vijayalakshmi B., **Kaliappan M.**, Ramar K., Vijayalakshmi K., Vimal S. (2021) Forecasting Using Deep Learning Approaches. pp 167-186. In: Srivastava S., Khari M., Gonzalez Crespo R., Chaudhary G., Arora P. (eds) Concepts and Real-Time Applications of Deep Learning. EAI/Springer Innovations in Communication and Computing. Springer, Cham. https://doi.org/10.1007/978-3-030-76167-7_11, Online: 24 September 2021.
5. S. Vimal, Y. Harold Robinson, **M. Kaliappan**, K. Vijayalakshmi, Sanghyun Seo, “Progression Detection of Glaucoma Using K-means and GLCM Algorithm”,In: Arabnia H.R., Ferens K., de la Fuente D., Kozerenko E.B., Olivas Varela J.A., Tinetti F.G. (eds) Advances in Artificial Intelligence and Applied Cognitive Computing. Transactions on Computational Science and Computational Intelligence. Springer , pp 863-868|, 2021.
6. P. Subbulakshmi, S. Vimal, **M. Kaliappan**, Y. Harold Robinson, Muccheol Kim, “Trend Analysis Using Agglomerative Hierarchical Clustering Approach for Time Series Big Data”,In: Arabnia H.R., Ferens K., de la Fuente D., Kozerenko E.B., Olivas Varela J.A., Tinetti F.G. (eds) Advances in Artificial Intelligence and Applied Cognitive Computing. Transactions on Computational Science and Computational Intelligence. Springer , pp 869-876|, 2021

5 Funded Projects

SNo	Title of the project	Funding Agency	Duration	Amount	Status
1	Automatic Book Reading System for Visually Impaired Person Using Deep Learning	Institution of Engineers-India(IE(I))	0.6 year	0.20 lakhs	Completed

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6 Short term Courses / Seminars/Webinars /FDP/Workshop/ConferenceAttended:

Sl.No	Title of the event	Organized by	Date	
			From	To
1.	Bridge 2023 - Madurai (A High Impact Industry-Institute Interaction Event of India)	Organized by ICT Academy held on at Courtyard by Marriott, Madurai	-	30.09.2023
2.	Webinar on Research Proposal Writing	Ramco Institute of Technology, Rajapalayam	30.01.2023	30.01.2023
3.	Machine Learning in the age of AI: Algorithms, Models and Beyond”	KPR Institute of Engineering and Technology Coimbatore	21-08-2023	01.09.2023
4.	Anna University Sponsored FDP on Artificial Intelligence and Machine Learning	National Engineering College, Kovilpatti	06.02.2023	11.02.2023
5.	Computational Techniques and Research Methodology	UGC sponsored Jawaharlal Nehru University, New Delhi	7.03.2022	12.03.2022
6.	Webinar on Safeguarding World Trade Markets through Analytics, AI, and ML on Cloud	TECHGIG	15.12.2022	15.2.2022
7.	Ethical Hacking	National Institute of Technical Teachers Training and Research (NITTTR) Chandigarh. Cyber Security Project Team, MeITY	28.06.2021	30.06.2021
8.	Workshop on Tableau Data Analysis	AICTE	05.07.2021	06.07.2021
9.	Data Analysis using Deep learning	National Engineering College, Kovilpatti	14.12.2020	19.12.2020
10.	Cypher Physical Systems	ATAL	07.11.2020	11.11.2020
11.	Incorporating Universal Human Values in Technical Education	AICTE	28.07.2020	01.08.2020
12.	Data Analytics using python	ICT Academy, Government of India	18.05.2020	23.05.2020
13.	Cyber Security Era in Artificial Intelligence	Bennet University, Noida	13.05.2020	17.05.2020
14.	Exam Reforms	AICTE, New Delhi	29.04.2020	02.05.2020

15.	GIAN course on Physics and biology inspired optimization, machine learning, data mining techniques and their applications in big-data, medical, science and engineering disciplines	IIT Bhubaneswar, Odisha	12.05.2019	23.05.2019
16.	International Engineering Educators Certification Programme Phase – I Workshop	Indo – Universal Collaboration for Engineering Education (IIUCEE) and Thiagarajar College of Engineering, Madurai	01.04. 2019	03.04.2019
17.	Conclave on Engineering & Technology	Thiagarajar College of Engineering, Madurai	14.03.2019	-
18.	ABCD: An Algorithmic Toolkit for BigData and e- Commerce	IIT Madras, Chennai	17.10.2016	28.10.2016
19.	National Cyber defense summit	Anna University Chennai	30.10.2016	01.11.2016
20.	Formulation of Research and Development Grant proposals by scientist and technologists	National Engineering College, Kovilpatti	18.11.2015	20.11.2015
21.	Intellectual property rights & Innovations	National Engineering College, Kovilpatti	07.11.2015	-
22.	NPTEL workshop	National Engineering College, Kovilpatti	24.04.2015	-
23.	ISTE convention	National Engineering College, Kovilpatti	07.11.2014	08.11.2014
24.	Train the Trainer(TTT) on Mobile Application Development using Android	Prathyusha Institute of Technology, Chennai	01-04-2014	03-4-2014
25.	Hands on Seminar on LATEX for Beginners	National Engineering College, Kovilpatti	14-6-2013	15-6-2013
26.	.NET Programming	National Engineering College, Kovilpatti	13.02.2012	20.02.2012
27.	.NET Programming	APC College of Arts and Science, Tuticorin	13.02.2012	20.02.2012
28.	Research Issues in Cloud Computing and Its Simulation	Mepco Engineering College Sivakasi	12.09.2012	13.09.2012
29.	Media Security : A set- theoretic framework for data hiding	National Engineering College Kovilpatti	29.08. 2012	-
30.	FOSS	Mepco Shelenk Engineering College, Sivakasi	29.09.2010	30.09.2010
31.	Wireless adhoc and sensor network	IIT, Kharagpur	02.02.2010	15.02.2010

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32.	Network Programming	C-DAC, Hyderabad	05.12.2008	12.12.2008
33.	Autonomic Computing	National Engineering College, Kovilpatti	01.08.2008	02.08.2008
34.	Web Engineering & e-Biz solutions	Mepco Shelenk Engineering College, Sivakasi	28.10.2007	08.11.2007
35.	J2EE and Web sphere Application Developer	Arulmigu kalasalingam college of Engineering, Srivilliputtur.	09.08.2007	10.08.2007

7. Short term Courses / Seminars /FDP/Workshop/Conference organized:

Sl.No	Title of the event	Organized by	Date	
			From	To
1.	ICSSR Sponsored Seminar on "Explainable Artificial Intelligence (XAI) inspired Learning approaches using Data Analytics for Education System	Ramco Institute of Technology, Rajapalayam	23.08.2023	25.08.2023
2.	Five-day Faculty Development Programme on Data Analytics and Machine Learning	Ramco Institute of Technology, Rajapalayam	13.02.2023	17.02.2023
3.	Technical seminar on Carrier path in Computer Network: Advanced Networking Technology & Physical layer	Ramco Institute of Technology, Rajapalayam	09.05.2023	09.05.2023
4.	Guest lecture on AWS cloud Deployment and AI Service	Ramco Institute of Technology, Rajapalayam		20.10.2023
5.	Webinar on Metaverse and its Applications	Ramco Institute of Technology, Rajapalayam	-	07.10.2023
6.	Virtual tech talk on Latest Developments in AI and its Application	Ramco Institute of Technology, Rajapalayam	-	15.09.2023
7.	Webinar on AI & ML in Today's World	Ramco Institute of Technology, Rajapalayam	-	18.09.2023
8.	Webinar on Generative AI and No-code platform roles in Industry	Ramco Institute of Technology, Rajapalayam		26. 08.2023
9.	Webinar on AI powered Data Analytics	Ramco Institute of Technology, Rajapalayam	-	19.08.2023
10.	SAP in Enterprise Networks and its opportunities	Ramco Institute of Technology, Rajapalayam	-	04.03.2023
11.	Guest Lecture Software Engineering an Insider View	Ramco Institute of Technology, Rajapalayam	-	18.12.2022

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12.	Webinar on Project Management and JIRA tool	Ramco Institute of Technology, Rajapalayam	-	13.12.2023
13.	motivational talk on Industry trends and Interview preparation	Ramco Institute of Technology, Rajapalayam	-	5.12.2022
14.	Webinar on Software Industry Trends in Artificial Intelligence and Data Science	Ramco Institute of Technology, Rajapalayam	-	02.11.2022
15.	Seminar on Exposure to the Industry Standards of Web3, Blockchain and AI/ML	Ramco Institute of Technology, Rajapalayam	24.09.2022	24.09.2022
16.	Seminar on Data lake and analytics in Amazon Web service	Ramco Institute of Technology, Rajapalayam	17.09.2022	17.09.2022
17.	workshop on Data Science	Ramco Institute of Technology, Rajapalayam	13.08.2022	13.08.2022
18.	Trend and Opportunities of Artificial Intelligence and Data Science	Ramco Institute of Technology, Rajapalayam	-	25.08.2022
19.	Induction program on Cutting Edge Technologies & React Framework	Ramco Institute of Technology, Rajapalayam	-	12.08.2022
20.	Webinar on CNN and Applications	Ramco Institute of Technology, Rajapalayam	-	01.02.2022
21.	Advance Data structure and its Application, Minimum Spanning Tree, Shortest path Algorithm & Dynamic Programming.	Ramco Institute of Technology, Rajapalayam	-	18.06.2022
22.	Workshop on Tour of Arduino	Ramco Institute of Technology, Rajapalayam	-	04.05.2022
23.	Workshop on Web Development	Ramco Institute of Technology, Rajapalayam	-	15.02.2022
24.	National Summer Training Program on Big Data and Hadoop	National Engineering College, Kovilpatti	27.06.2016	03.07.2016
25.	Anna University Chennai sponsored FDTP on Web programming	National Engineering College, Kovilpatti	06.06.2016	12.06.2016
26.	DSTNIMAT sponsored Entrepreneurship Awareness Camp	National Engineering College, Kovilpatti	19.01.15	21.01.15
27.	DBT sponsored National level seminar on Research approach and contemporary issues of functional genomics in Bioinformatics application	National Engineering College, Kovilpatti	13.11.14	13.11.14

28.	DBT sponsored on Research & Pragmatic Issues of Cloud Computing Genomic Analysis in bioinformatics Applications”	National Engineering College, Kovilpatti	14.03.13	15.03.13
29.	Anna University sponsored FDTP on Service Oriented Architecture	National Engineering College, Kovilpatti	9.06.14	15.06.14
30.	DSTNIMAT sponsored Entrepreneurship Awareness Camp”	National Engineering College, Kovilpatti	22.01.14	24.01.14
31.	National Conference on ENIAC’12	National Engineering College, Kovilpatti	12.10.12	
32.	Workshop on NS2	National Engineering College, Kovilpatti	26.09.2014	27.09.2014
33.	Workshop on NS2	National Engineering College, Kovilpatti	04.10.13 to 05.10.13	10.10.13 to 11-10-13

8. Expert Lectures Delivered :

Sl. No	Topic	Date	Organization
1.	Three day workshop on LMS-Canvas	08.06.2020 to 10.06.2020 (9.30 pm to 4.00 pm)	Dept. of CSE, RIT Participants: Ramco ITI faculty, No. 18
2.	Webinar on Deep learning	06.06.2020 (11.00 am to 12.30 pm)	Jeyaraj Annapackiyam CSI college of Engineering Nazareth
3.	Webinar on LMS-Canvas	20.05.2020 to 22.05.2020 (2.00 pm to 3.00 pm)	PACR Poly technique College, Rajapalayam
4.	Seminar on Teaching Learning process	4.11.19	Ramco Institute of Technology
5.	Seminar on Course Design and dynamic class room	6.11.19	Ramco Institute of Technology
6.	Value added course on website design and hosting	27.11.2019 28.11.2019	Ramco Institute of Technology
7.	Workshop on Educational Technology for Gen-Z students	16.04.2019	Ramco Institute of Technology
8.	Finishing School program-Python programming	24.04.2019	Ramco Institute of Technology
9.	Value added course on Problem solving Techniques	17.06.2019-20.06.2019	Ramco Institute of Technology,
10.	Value added course Projects using PHP	24.06.2019-26.06.2019	Ramco Institute of Technology
11.	Placement Training- Mock Interview	27.06.2019	Ramco Institute of Technology,
12.	FDTP on Service Oriented Architecture	09-06-2014- 15-06-2014	National Engineering college, Kovilpatti
13.	One day workshop on Android application development	09.10.2014	JACSI college of Engineering Nazareth

14.	Workshop on Android and its Applications	02.08.2014	IEEE Student branch , National Engineering college, Kovilpatti
15.	Workshop on NS2	26.09.14 to 27.09.14	National Engineering college, Kovilpatti
16.	Workshop on .NET	30-12-2013	G.Venkataswamy Naidu College,Kovilpatti
17.	Wireless Network Simulator using NS2	4-10-2013 - 5-10-2013	National Engineering college, Kovilpatti
18.	Workshop on J2EE	26.09.2009	National Engineering college, Kovilpatti

9. Online Courses

sno	Title	University Offered By	Duration	Date	Completed on
1	Linear Algebra - Foundations to Frontiers	University of Texas System	2 months	20.08.2023	22.10.2023
2	Tableau Desktop from Scratch –Data Analysis and Visualization	Udemy	1 day	10.10.2022	10.10.2022
3	Time series Data Analysis	Udemy	1 day	10.11.2022	10.11.2022
4	Python for Deep Learning: Build Neural Networks in Python	HarvardX	1 day	12.04.2022	12.04.2022
5	Data Manipulation in Python: Master Python, Numpy & Pandas	HarvardX	1 day	12.04.2022	12.04.2022
6	Python for Data Analysis & Visualization Masterclass 2022	HarvardX	1 day	07.04.2022	07.04.2022
7	Data Science and Machine Learning Capstone Project	IBM	5 weeks		30.08.2020
8	Data Science: Probability	HarvardX	8 weeks	01.07.2020- 30.08.2020	30.08.2020
9	Computer Vision Fundamentals with Watson and OpenCV	IBM	14 weeks	15.04.2020- 30.08.2020	30.08.2020
10	Data Science Tools	IBM	7 weeks		30.08.2020
11	SQL for Data Science	IBM	4 weeks		30.08.2020
12	Machine Learning with Python: A Practical Introduction	IBM	5 weeks		28.08.2020
13	Deep Learning with Tensor flow	IBM	5 weeks		10.08.2020
14	Introduction to Data Science	IBM	6 Weeks		22.07.2020
15	Visualizing Data with Python	IBM	5 weeks		27.07.2020
16	Analyzing Data with Python	IBM	5 Weeks	02.06.2020 20.07.2020	20.07.2020
17	Python Basics for Data Science	IBM	1 week		10.07.2020

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18	Data Science: Linear Regression	HarvardX	8 weeks	01.06.2020-30.07.2020	29.07.2020
19	Calculus 1C: Coordinate Systems and Infinite Series	Massachusetts Institute of Technology	8 weeks		12.05.2019
20	International Engineering Educator Certification Program	Indo Universal Collaboration for Engineering Education	6 months		22.11.2019
21	Data Analytics	NPTEL	12 weeks		2016
22	Introduction to Research	NPTEL	12 weeks		2017

10. Patents filed/awarded

SN o	Application No/type	Title	Inventors	Applicant name	Date of Filing application	Status
1	2020103425 International Country: Australia	An Artificial Intelligence based Solar Powered Glove with Gestures for Deaf and Dumb	Dhiman, Gaurav , Gomathi, S. , Arora, Vinay ,Singh, Krishna Kant ; Vimal, S., Kaliappan, M. , Singh, Akansha , Soni, Mukesh, Kaur, Amandeep , Saxena, Deepika		13.11.2020	Published On 13.01.2021
2	202041046216 Indian	Deep learning based Yoga Pose Assessment System	1.Dr.L.Ganesan 2.Dr.K.Vijayalakshmi 3. Dr.M.Kaliappan 4. R.Venkatesh 5.K.Vignesh saravanan 6.Kata Bhatat	Ramco Institute of Technology	23/10/2020 30/10/2020	Filed
3	Application no: 10-2020-0099721, Patent office number: P20187 International patent: Korean patent	Drone detection apparatus and method using deep learning	Mi Young Lee , M. Kaliappan , S Vimal, K. Vijayalakshmi , S. Manikandan	Applicant name: Sejong University Industry-Academic Cooperation Foundation, Korea	10.08.2020	Filed
4	201941038317 A Indian	Global System For Mobile Communication (GSM)	1.A Suresh 2.S.Vimal 3. N.Malarvizhi 4. M.Kaliappan 5.P.Subbulakshmi 6.Neeba E A 7. S. Sasikala		23/09/2019	Published Dt. 25/10/2019

11. Session chair

SNo	Title of the Event	Organized by	Topic	Role , Date
1	World Automation Congress (WAC) 2020 Technical Sponsor: IEEE SMC Society	Conference Center, Taipei Tech Taipei, Taiwan	Energy conservation and enhancement in wireless cognitive radio networks with Analytics	Session Organiser Aug 16-20, 2020
2	2 nd International Conference on Emerging Technologies in Data Mining and Information Security. Sponsored by Springer	Institute of Engineering & Management Group Technically Sponsored by The Smart Society, USA. Venue: Institute of Engineering & Management (Ashram Building) , Kolkata	Session ID : IEMIS/2020/3 Applications of Game theory to enhance the security and energy monitoring for Cognitive radio Networks	Co-Chair 2nd - 4th July, 2020
3	IEEE 8 th Global Conference on Consumer Electronics (GCCE 2019)	5F and 6F, Senri Life Science Center. 1-4-2, Toyonaka-city, Osaka, Japan.	Data Analytics and Security Enhancement with Applications of Game Theory for Cognitive Radio Networks	Co-Chair October 15-18, 2019
4	International Conference on Recent Trends in Information and Communication Engineering	Jayaraj Annapackiam CSI College of Engineering, Nazareth, Tamil Nadu 628617	Paper presentation	Advisory committee member 08.03.2019
5	State Level Technical Symposium-TECHYUGA-18	Ramco Institute of Technology, Rajapalayam	Paper presentation	Chairperson 31.08.2018
6	National Level Technical Symposium Watts on 2K18	Kalasalingam University, Srivilliputtur, Virudhunagar District	Paper presentation	Chairperson 31.03.2018

11. Membership in Professional societies

- 1) Institution of Engineers (India) Limited, Kolkata, India. Life Member - (M-163955-7)
- 2) CSI member

12. Awards and Achievements

- Nominated as **reviewer** for International Conference on Emerging Technology and Interdisciplinary Sciences (ICETIS)-2021 on 25-26 June, 2021
- Received **letter of Appreciation** ensuring a smooth completion of the various academic activities in the COVID-19 impacted semester during Jan-April 2020
- **Editorial members** in International Journal of Advanced Science and Information Technology, War Publishers. <http://warpublishers.com/journals/IJASIT/editorial.html>.
- **Editorial member** in *Journal of Networking and Telecommunications*, Pisco med publishing, ISSN: 2661-4065, <http://ojs.piscomed.com/index.php/JNT/about/editorialTeam>
- Appreciation for **Active NPTEL-SPOC** based on performance and Participation of candidates for the JAN-APR 2019 semester.
- Recognized as **Supervisor for Ph.D/MS** programme in Anna university
- Received appreciation for **Active NPTEL-SPOC** based on performance and Participation of candidates for the Jan-Apr., 2019
- completed **IUCEE International Engineering Educators Certification Program (IIEECP) with distinction**
- Acted as **Panel member** in Virtual Conference Organizers' Panel organized by **IEEE Bangladesh section**, Bangladesh.
- Acted as **session Co-Chair** in IEEE 8th Global Conference on Consumer Electronics (GCCE 2019) on Data Analytics and Security Enhancement with Applications of Game Theory for Cognitive Radio Networks, organized by Senri Life Science Center, Toyonaka-city, **Osaka, Japan**.
- Acted as **session chair** 2nd International Conference on Emerging Technologies in Data Mining and Information Security, organized by Institute of Engineering & Management Group, **Kolkata, Technically Sponsored by The Smart Society, USA** and Sponsored by Springer.
- Doctoral Committee member for a Ph.D Scholar of Anna University, Chennai
- Acted as Oral panel member for Scholar of Anna University, Chennai
- Act as observer for central teacher eligibility test at Madurai on 08.12.2018

Others (Board of studies/Governing Council/Staff Selection/ Reviewer/other duties)

- **Reviewed papers** for IEEE Access Journal, IET Networks, Mobile and Pervasive computing